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AGRI CATALOGUES
www.agricatalogues.in

ISSN: 3108-124X

Powdery Mildew of Cucurbits: A Persistent Threat to Vegetable Production

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Cucurbitaceous vegetables such as cucumber, pumpkin, squash, melon, watermelon, bottle gourd and other gourds are cultivated widely across tropical, subtropical and temperate regions. These crops play a vital role in human nutrition and farm income. However, their production is severely constrained by several fungal diseases like Phytophthora Blight, Downy Mildew, Powdery Mildew, damping off and Choanephora Blight, among which Powdery Mildew is considered one of the most destructive and recurrent foliar disease, caused by two fungal pathogens, *Podosphaera xanthii and Erysiphe cichoracearum*. The disease is easily recognisable by the characteristic white, powdery fungal growth on leaves and stems and is capable of causing serious yield and quality losses if not managed properly (McGrath, 1996).

Powdery mildew occurs in both open fields and protected cultivation conditions and can infect plants at almost any growth stage. Unlike many other fungal diseases, it does not require free water for infection, which allows it to flourish even under relatively dry conditions with moderate humidity (Agrios, 2005).

Keywords: Powdery mildew, Cucurbits, Fungal diseases, Fungi, Fungicides and Azoxystrobin 23% SC.

Economic Consequences

Powdery mildew is responsible for significant yield losses, sometimes ranging from 20–50% in susceptible cultivars under favourable conditions. Early defoliation reduces fruit filling and shortens the productive life of the crop. In crops like muskmelon and cucumber, the disease can drastically reduce marketable yield and consumer acceptability (McGrath and Thomas, 1996).

Pathogenic Organisms Involved

Powdery mildew of cucurbits is caused mainly by two fungi, *Podosphaera xanthii*(earlier known as *Sphaerotheca fuliginea*) and *Erysiphe cichoracearum.* Among these, P. *xanthii* is the most common and damaging species in cucurbit fields across Asia, Europe and the Americas (McGrath, 1996). These fungi are obligate parasites, which means they can survive and multiply only on living cucurbit plants. Once established on the crop, they spread rapidly through airborne spores, making early detection and timely management essential for farmers to avoid heavy yield losses.

Host Range and Distribution

Powdery mildew can infect almost all cucurbitaceous crops, but susceptibility varies among species and varieties. Crops like cucumber, pumpkin, squash, muskmelon, bottle gourd and ridge gourd are highly vulnerable, whereas watermelon and some resistant cucumber varieties tend to be less affected (McGrath, 1996; Pawar and Chavan, 2010). This means that even if a farmer grows multiple cucurbits in the same field, some crops may show severe infection while others remain relatively healthy.

The disease is found worldwide, in both tropical and temperate regions, making it a constant threat throughout the growing season. In India, powdery mildew diseases of different crops are most commonly observed during the Rabi and summer seasons, especially when days are warm and nights are cool, conditions that favour spore germination and rapid disease spread (Pawar and Chavan, 2010). Farmers should therefore

monitor their fields closely during these seasons, as early detection and timely management are crucial to prevent heavy leaf damage and significant yield loss.

Characteristic Symptoms

Powdery mildew often begins on the older leaves of the plant. Farmers may first notice small, pale yellow spots on the upper surface of these leaves. Within a few days, these spots develop into white or greyish powdery patches, which are actually fungal growth and spores. If not controlled, the powder spreads quickly, covering entire leaves, leaf stalks (petioles) and sometimes stems (Zitter *et al*., 1996).

As the disease progresses and becomes severe, several problems appear:

- **Leaf damage:** Leaves turn yellow, then brown, dry up and may fall off earlier than normal.
- **Weaken plants:** Continuous loss of leaves weakens the plant, reducing growth and vigour.
- **Fruit exposure:** Loss of foliage exposes fruits to direct sunlight, leading to sunburn (sunscald).
- **Reduction in fruit quality:** Even though fruits are rarely infected directly

Figure 1 - Affected parts of Cucurbits by Powdery Mildew



Healthy leaves are crucial for producing energy through photosynthesis. When powdery mildew damages leaves, the plant is unable to produce enough food, resulting in smaller fruits and lower yields. Early detection and management are therefore critical to protect both the plant and the harvest.

Epidemiology and Climatic Regulation

Powdery mildew usually appears first on the older leaves of cucurbit plants as small, pale yellow spots on the upper surface, which quickly develop into white or grey powdery patches of fungal growth and spores (Zitter *et al*., 1996). In severe cases, leaves turn yellow and brown, dry up and may fall off prematurely, weakening the plant and reducing growth (McGrath, 1996). Fruits are rarely directly infected, but the disease can reduce fruit size, sweetness, colour and storage quality by lowering photosynthesis (McGrath, 1996).

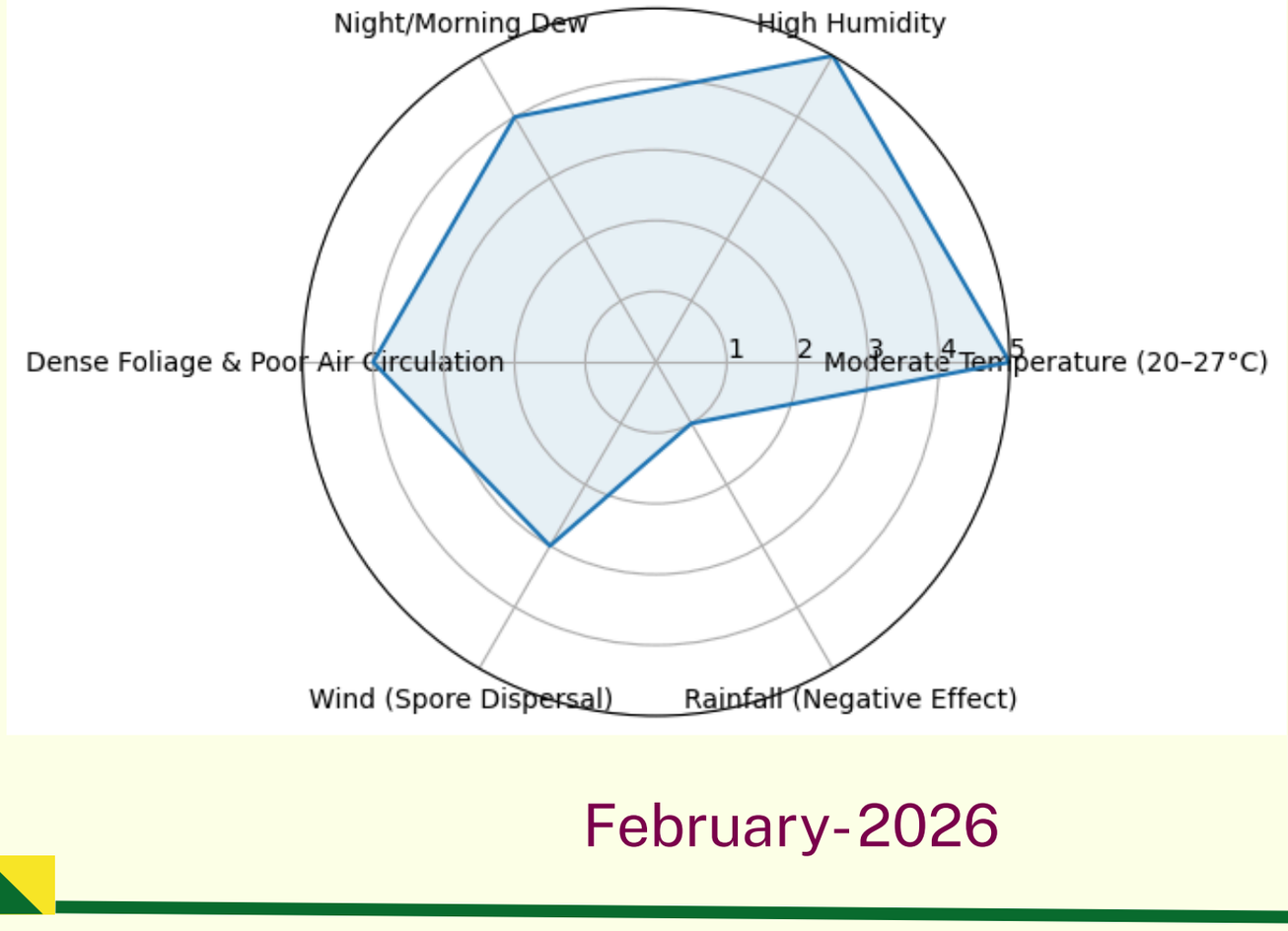


Figure 2 - Optimum conditions for Powdery Mildew infection

The disease develops most rapidly under moderate temperatures (20–27 °C) and high humidity, especially during nights or when morning dew is present (Agrios, 2005). Dense foliage and poor air circulation further favor infection, while wind helps spread spores over long distances (Agrios, 2005). Unlike many fungal diseases, powdery mildew does not require water on the leaves to infect, heavy rainfall may even reduce disease severity by washing away spores (Agrios, 2005).

Disease Development

Powdery mildew survives between cropping seasons mainly on plant debris as sexual fruiting bodies called cleistothecia, or on living cucurbit plants and certain weeds in mild climates. During the growing season, the fungus produces large numbers of asexual spores known as conidia, which are easily carried by the wind to healthy plants. Once on a leaf, these spores germinate and penetrate the surface using specialized structures called haustoria, establishing infection. The fungus can complete several cycles of spore production and infection in a single season, allowing the disease to spread rapidly and sometimes cause severe outbreaks under favourable conditions (Zitter *et al.,* 1996).

Management Strategies

Effective management of powdery mildew relies on an integrated disease management approach combining cultural practices along with chemical fungicides.

Cultural practices

One of the most economical and environmentally safe ways to manage powdery mildew is the use of resistant varieties. Proper plant spacing also plays a key role, as it improves air circulation and reduces humidity within the crop canopy, making conditions less favourable for the fungus. Removing infected plant debris after harvest is important because it helps reduce the amount of fungus that can survive between seasons. Additionally, balanced fertilization is essential, avoiding excessive nitrogen prevents overly lush growth, which can create a humid microclimate that encourages disease development (Agrios, 2005). Analysis of the five mature leaves for powdery mildew infection (once a week) could be done too for effective control of the disease (Nama *et al*., 2016).

Chemical control

Fungicides play a crucial role in the management of powdery mildew, particularly under intensive and commercial cucurbit cultivation systems. Preventive applications made before the appearance of visible symptoms are generally more effective than curative sprays applied after disease establishment. To minimize the risk of fungicide resistance development in the pathogen population, repeated use of fungicides with the same mode of action should be avoided and rotation of fungicides with different modes of action is strongly recommended (McGrath, 1996).

In cucumber, powdery mildew can be effectively managed by spraying Azoxystrobin 23% SC @ 500 ml per hectare diluted in 500 litres of water, whereas in watermelon, effective disease control can be achieved through spraying Bupirimate 26.7% w/w EC @ 2000 ml per hectare in 500 litres of water. In addition, powdery mildew of cucurbits can also be managed by spraying Carbendazim 50% WP @ 300 g per hectare diluted in 600 litres of water. In addition to these fungicides, powdery mildew of cucurbits can also be effectively managed by spraying Metrafenone 500 g/L SC @ 250 ml per hectare diluted in 500 litres of water, which provides good preventive and curative control.

Under organic farming systems, sulfur-based fungicides and plant-derived formulations (botanicals) are commonly used for powdery mildew management. Although these treatments can provide satisfactory disease suppression, caution is required during application, particularly under high temperature conditions, to avoid phytotoxic effects on foliage.

Integrated management system

An integrated disease management (IDM) approach offers the most sustainable and reliable control of powdery mildew in cucurbit crops. This strategy combines the use of resistant cultivars, proper cultural practices such as adequate spacing and balanced fertilization, regular field monitoring for early disease detection and the careful use of fungicides only when necessary. By integrating these methods, disease pressure is reduced, the risk of fungicide resistance is minimized and dependence on chemicals is lowered. IDM not only helps protect the crop throughout the growing season but also reduces production costs and environmental impact, making it a practical and eco-friendly approach for farmers.

Conclusion

Powdery mildew of cucurbits is a persistent and economically important disease that affects cucurbit production worldwide. Its unique ability to thrive without free water and spread rapidly through wind-borne spores makes its management difficult. However, with early detection, resistant varieties and integrated management practices, the disease can be effectively controlled. Awareness among farmers and students is essential for timely intervention and sustainable cucurbit cultivation.

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